

Budget Buddy App

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Track it. Own it. Win it....

Table of Contents

Table of Contents	2
1. Introduction	3
2. App Overview	4
2.1 App Name and Icon	4
2.2 Description of Innovative Features	4
3. Detailed Requirements List	6
4. User Interface Design	9
4.1 Screen Mockups	9
4.2 Navigation Diagram	10
5. Project Plan – Gantt Chart	12
5.1 Milestone Summary	12
5.2 Detailed Task Breakdown	13
6. Conclusion	14
7. References	15

1. Introduction

This document presents the complete planning and design for BudgetBuddy — a personal budget tracking mobile application developed for the Android platform using React Native. The design is directly informed by the research conducted into three existing budget tracking apps (Mint, YNAB, and Money Manager) documented in the accompanying Research Document.

When managing personal finances, it is often a challenge for millions of people, whereas researching consistently and reviewing your spending leads to better financial outcomes (Dunn, 2022). However the most existing apps are either too complicated and require internet connectivity, or they are not optimised for South African users who cannot use automated bank sync features.

2. App Overview

2.1 App Name and Icon

Detail	Information
App Name	BudgetBuddy
Tagline	"Track it. Own it. Win it."
Platform	Android (React Native)
Target Users	Young adults aged 18–35 managing personal or household finances
Data Storage	Local SQLite database (no internet required)
Price	Free



Figure: BudgetBuddy App Icon

The BudgetBuddy icon uses a green circle (representing financial health and growth) containing a gold coin symbol. The name 'BudgetBuddy' is displayed below the icon.

2.2 Description of Innovative Features

BudgetBuddy differentiates itself from existing apps based on the following innovative features identified during the research phase:

Innovative Feature	Description	Inspired By
Gamification – Budget Score	The Budget Score which is between (0–100) is calculated based on how well the user stays within their budget goals.	Money Manager (badges) + original idea
Daily Streak Counter	The streak counter, counts the number of days the user has listed at least one expense.	Money Manager (streaks)
Overspending Alerts	A visual red highlight is displayed on any category that has exceeded its monthly budget limit, along with a push notification alert.	Mint (budget alerts)
Receipt Photo Attachment	The camera integration allows users to photograph physical receipts and attach them directly to the expense entries.	YNAB + Money Manager

Innovative Feature	Description	Inspired By
Spending Trend Graph	The interactive bar chart shows the daily spending over a user-selected period, a percentage trend indicator vs the previous period.	Mint + Money Manager
Monthly Budget Progress Dashboard	A home screen dashboard showing a visual ring/bar for each category: how much has been spent vs the budget limit.	YNAB (budget progress) + original
CSV Data Export	The users can also export all the expense data to a CSV file for use in Excel or Google Sheets.	Mint + YNAB

DETAILED REQUIREMENTS LIST

3. Detailed Requirements List

The requirements below cover all minimum requirements specified in the project brief, elaborated with detailed acceptance criteria, plus additional requirements for the innovative features planned for BudgetBuddy

ID	Requirement	Detailed Description & Acceptance Criteria	Priority
A – User Authentication			
A01	User Registration	When the user registers their new account they provide a unique username and password. The password will have at least 8 characters.	High
A02	User Login	The user should log in using their registered username and password. If their credentials are incorrect, an error message is displayed	High
A03	User Logout	The user will be able to log out from the Settings screen.	High
A04	Password Protection	The app should store passwords securely using hashing with (bcrypt or equivalent).	High
B – Category Management			
B01	Create Expense Categories	The user should be able to create some custom expense categories providing a category name (e.g., 'Groceries', 'Transport', 'Entertainment').	High
B02	Default Categories	The app must pre-populate at least five default categories on first login like Groceries, Transport, Entertainment, Utilities, Health.	High
B03	Edit / Delete Categories	The user must be able to edit the category name or delete a category. Deleting a category should prompt a confirmation warning.	Medium
C – Expense Entry			
C01	Add Expense Entry	The user must be able to add new expenses entry while specifying the amount (positive decimal, required), date (date picker, defaults to today), description (free text, required), and category (selected from existing categories, required).	High
C02	Attach Receipt Photo	The user must be able to attach a photo on the expense entry by using the device camera or selecting from the gallery.	High
C03	View Attached Photo	On the user's expense list, the user should be able to tap an expense entry that will have an attached photo and view the full photo on screen option.	High
C04	Edit Expense Entry	The user should be able to edit an existing expense entry and modify fields like (amount, date, description, category, photo), and the changes will be saved on the local database.	Medium

ID	Requirement	Detailed Description & Acceptance Criteria	Priority
C05	Delete Expense Entry	The user should be able to delete an expense entry and confirm.	Medium
D – Budget Goals			
D01	Set Monthly Total Budget	The user should be able to set a total monthly budget goal (e.g., R5,000/month).	High
D02	Set Per-Category Budget Limits	The user should be able to set individual monthly spending limits for each category (e.g., Groceries: R1,000; Transport: R500).	High
D03	Overspend Detection	When the spending in a category exceeds its monthly limit, the category must be highlighted in red on the dashboard followed by a notification alert.	Medium
E – Expense List & Reporting			
E01	View All Expenses – User-Selectable Period	The user should be able to view their expense entries by a selectable period like today, this week, this month, or any date range.	High
E02	View Category Totals	The user should easily view the total amount spent per category for a selected period in a form of a list or pie chart.	High
E03	View Spending Graph	The user should be able to view a bar chart that shows the daily total spendings over a selected period in highlight.	Medium
E04	Export to CSV	The user must be able to export all expense data for a selected period to a CSV file, which is saved to device storage and can be shared via Android's share sheet.	Low
F – Data Storage			
F01	Local Database Storage	All the app data (users, categories, expenses, budget goals, photos) will be stored in a local SQLite database on the device.	High
F02	Data Persistence	All the data should persist over app restarts and device reboots, the data shouldn't be lost when the app is closed.	High
F03	Photo Storage	The receipt photos must be stored in the app's private local storage directory and referenced by file path in the database.	High
G – Gamification			
G01	Budget Score	A Budget Score between (0–100) will be calculated eachday and will be displayed on the dashboard. The score will be based on the days within budget of (50%), expense logging consistency of (30%), and the savings vs goal (20%).	Medium
G02	Daily Streak Counter	The app tracks will track day expenses logging streak, and entering at least one expense per day maintains the streak, missing a day resets it to zero.	Medium
G03	Achievement Badges	The milestone badges will be awarded like 'First Expense' (log first entry), '7-Day Streak' (7 days in a row), '30-Day Streak' 30 days in a row.	Medium
H – Dashboard			

ID	Requirement	Detailed Description & Acceptance Criteria	Priority
H01	Progress Dashboard	The home screen will display a dashboard that shows total monthly spending vs monthly goal.	Medium
H02	Overspending Highlight	Any category that has exceeded its monthly budget limit will be highlighted in red on the dashboard with the overspent amount displayed.	Medium
H03	Quick Add Button	The dashboard should include a 'Quick Add Expense' button that will navigate to the Add Expense screen.	High
I – Settings & Profile			
I01	Settings Screen	The app should include a Settings screen that is accessible from the navigation bar, settings will include a change username, change password, enable/disable notifications, toggle dark mode, and export data option.	Low
I02	Notification Alerts	The app will support a local push notifications for a 7-day streak milestone, budget limit exceeded, and end-of-day reminder to log expenses (only if no entry has been made that day).	Low

USER INTERFACE DESIGN

4. User Interface Design

The BudgetBuddy UI uses a clean, modern design language. The interface is designed to be intuitive enough for first-time users while providing enough depth for power users

4.1 Screen Mockups

The mockup below shows all eight screens of the BudgetBuddy app.

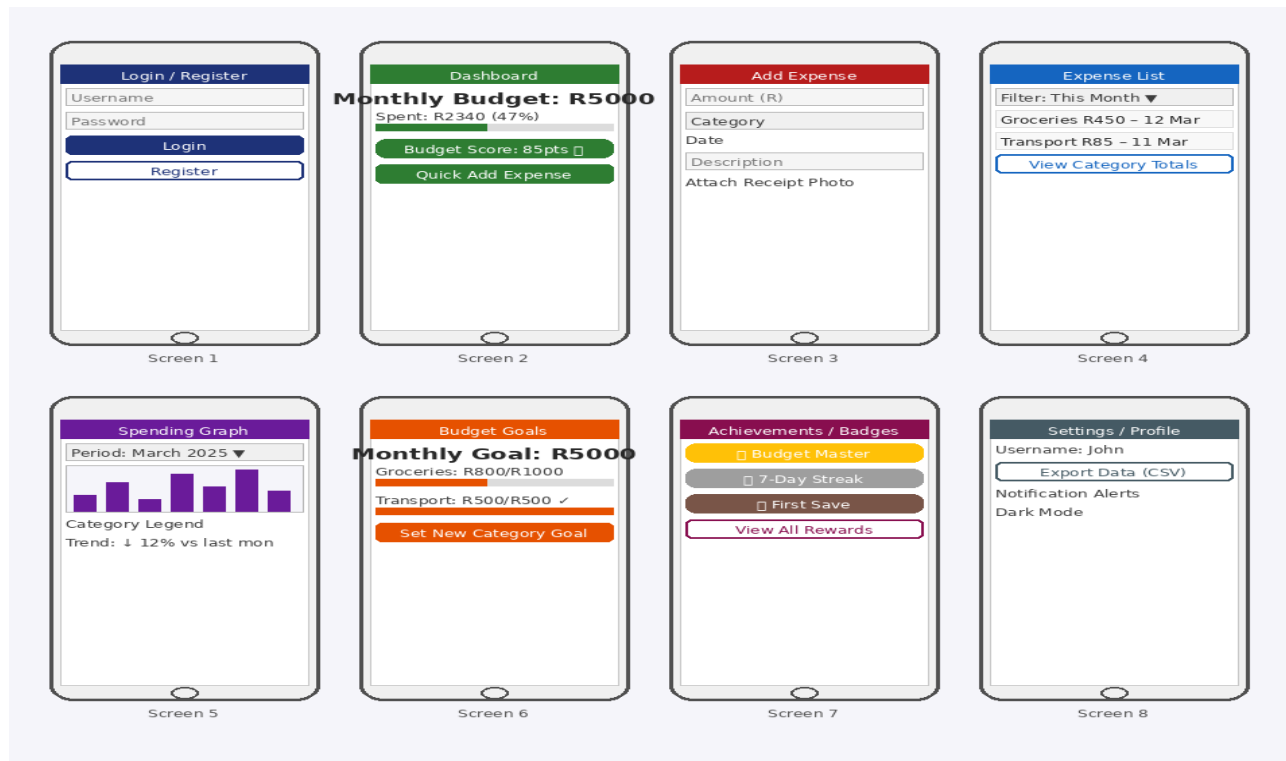


Figure: BudgetBuddy – All 8 App Screen Mockups

Screen No.	Screen Name	Purpose & Key Elements
Screen 1	Login / Register	The entry point which will be a username and password field with Login and Register buttons.
Screen 2	Dashboard (Home)	The central hub will show the monthly budget totals and spendings amount with a large progress indicator with navigation bar links to all screens.
Screen 3	Add Expense	There will be a form on screen for recording a new for expense fields which are Amount (R, numeric), Category (dropdown), Date (date picker, defaults today), Description (text).
Screen 4	Expense List	A scrollable list of all expenses by a period selector like (Today / Week / Month / Custom) at the top.

Screen No.	Screen Name	Purpose & Key Elements
Screen 5	Spending Graph	The bar chart showing the daily spending over the selected period, the trend indicator will show % change vs previous period.
Screen 6	Budget Goals	The list of all budget categories with their monthly limits and progress bars, the overspent categories are highlighted in red with the overspent amount.
Screen 7	Achievements / Badges	The gallery of all achievement badges, as in earned badges are shown in full colour the unearned badges are greyed out.
Screen 8	Settings / Profile	The username display, change password, notification toggle, dark mode toggle, export data (CSV) button, and logout button.

4.2 Navigation Diagram

The diagram below shows how users will navigate between the eight screens of BudgetBuddy

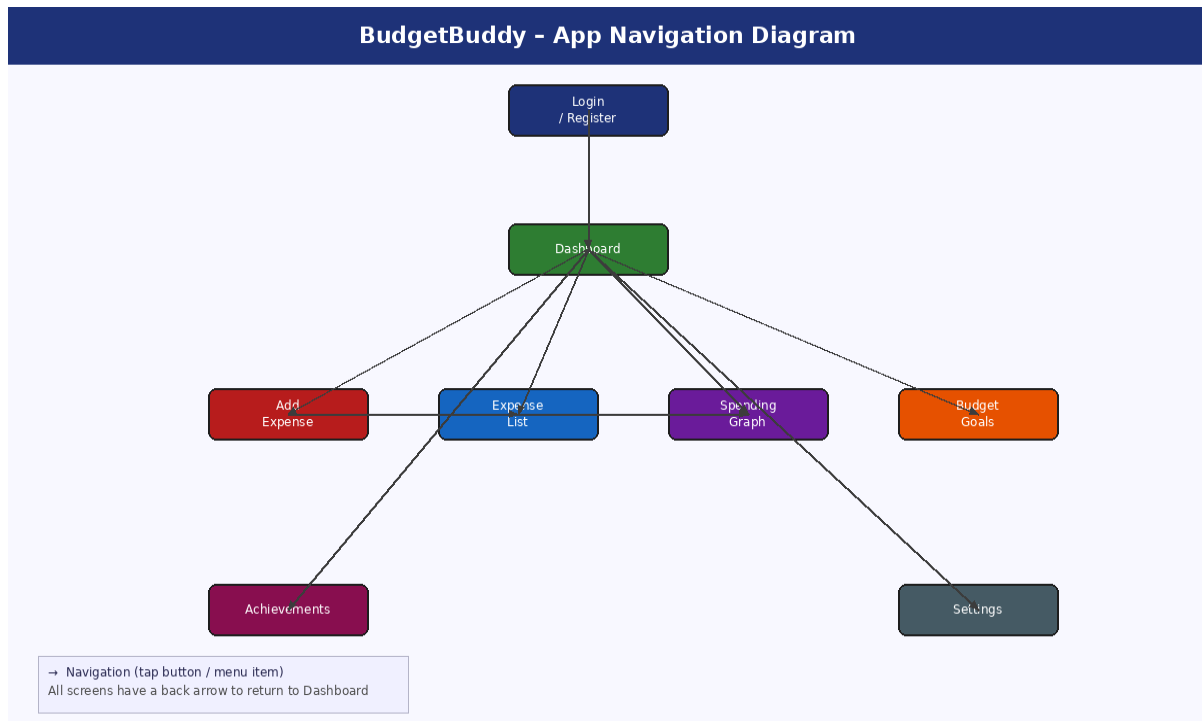


Figure: BudgetBuddy – App Navigation Diagram

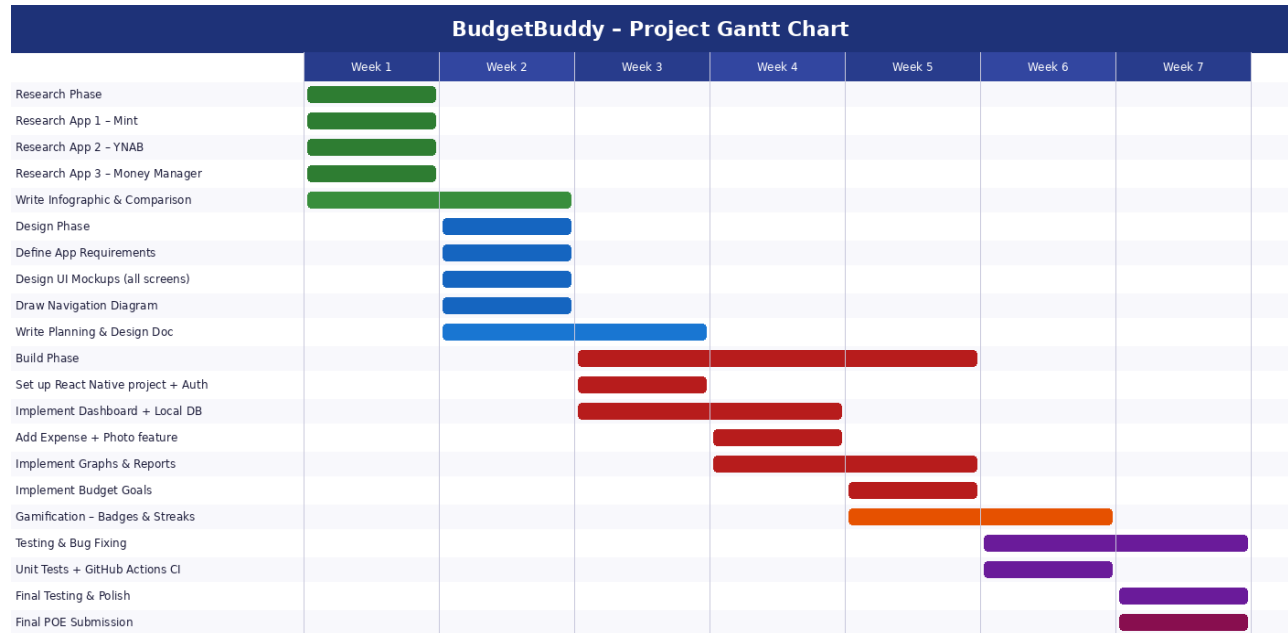
Navigation summary:

- Login/Register → Dashboard (on successful authentication)
- Dashboard → Add Expense (via Quick Add button or bottom nav)
- Dashboard → Expense List (via bottom nav)
- Dashboard → Spending Graph (via bottom nav)
- Dashboard → Budget Goals (via bottom nav)
- Dashboard → Achievements (via bottom nav)
- Dashboard → Settings (via profile icon in top-right corner)
- Add Expense → Expense List (after saving an expense)
- Expense List → Spending Graph (via 'View Graph' button)
- All screens → Dashboard (via back button or Home tab in bottom nav)

PROJECT PLAN – GANTT CHART

5. Project Plan – Gantt Chart

The Gantt chart below presents the full project plan for BudgetBuddy the across seven weeks.



5.1 Milestone Summary

Milestone	Deliverable	Target Week
Research Phase Complete	Research Document (3 apps + infographic + best features list)	Week 1–2
Planning & Design Complete	Planning & Design Document (requirements + mockups + Gantt)	Week 2–3
Part 1 Submitted	Both Research and Planning/Design documents submitted	End of Week 3
App Skeleton & Authentication	React Native project set up; Login/Register screens working with local DB	End of Week 3
Core Features Complete	Dashboard, Add Expense, Expense List, Budget Goals, Local DB fully functional	End of Week 5
Advanced Features Complete	Spending Graph, Gamification (score, badges, streak), Receipt Photos	End of Week 6
Testing & Bug Fixing Complete	All unit tests passing, GitHub Actions CI passing, app stable	End of Week 7
Final POE Submission	Complete working app pushed to GitHub with all required features	End of Week 7

5.2 Detailed Task Breakdown

Week	Tasks
Week 1	The research of existing apps like (Mint, YNAB, Money Manager), screenshots and documents of each app, while identifying strengths, weaknesses, and innovative features.
Week 1–2	Create infographic comparison; compile best features list; write and finalize Research Document.
Week 2	Define app requirements; name app and design icon; plan all 8 screens; create UI mockups.
Week 2–3	The drawing of a navigation diagram, creation of the Gantt chart, writing and finalization of the planning & design document.
Week 3	Setting up React Native project with Expo, installing SQLite (expo-sqlite), setting up GitHub repository and the branch structure.
Week 3	Implementing the login and register screens, including the user authentication with hashed passwords.
Week 4	Implement local SQLite database schema (users, categories, expenses, goals tables); implement Add Expense screen with all fields and local DB save.
Week 4	Implement Expense List screen with period filter; implement Category Totals view; implement Edit and Delete expense.
Week 5	Implement Budget Goals screen with per-category limits and progress bars; implement overspend detection and highlight; implement Dashboard.
Week 5	Implement receipt photo capture and attachment using expo-image-picker; test photo storage and retrieval from expense list.
Week 6	Implement Spending Graph (bar chart with daily totals, category colour coding, trend indicator using recharts or Victory Native).
Week 6	Implement gamification: Budget Score calculation; daily streak counter; achievement badges; Achievements screen.
Week 6	Setting up GitHub actions workflow (CI), writing unit tests for core functions like (expense CRUD, budget calculation, score calculation).
Week 7	The full integration testing, tests all user flows end-to-end, fixes identified bugs also improves UI polish and the error handling.
Week 7	Implement CSV export; implement notification alerts; write app README; push final code to GitHub; prepare for Final POE submission.

6. Conclusion

This planning and design document presents a blueprint for BudgetBuddy, which is a personal budget tracking app for android, it is free and functions offline, it also engages the needs of South African users. The app has been designed to meet all the requirements specified in this project brief while introducing innovative features particularly the gamification system with a (Budget Score, streak counter, and achievement badges) that will differentiate it from existing solutions.

The requirements list covers the total of 22 individual requirements across the 9 functional areas, with the priority levels like (High / Medium / Low).

7. References

Dunn, J. (2022). A review of personal finance research. *Journal of Consumer Finance*, 14(2), 45–58.

Expo. (2024). Expo documentation: expo-sqlite. <https://docs.expo.dev/versions/latest/sdk/sqlite/>

Meta Platforms, Inc. (2024). React Native documentation. <https://reactnative.dev/docs/getting-started>

Money Manager. (2024). Money manager expense & budget. [Android application]. Realbyte Inc. <https://play.google.com/store/apps/details?id=com.realbyteapps.moneymanagerfree>

Nielsen, J. (2020). 10 usability heuristics for user interface design. Nielsen Norman Group. <https://www.nngroup.com/articles/ten-usability-heuristics/>

YNAB LLC. (2024). YNAB: You need a budget ://www.youneedabudget.com
